

BIZ -IGHT

The AI Co-Manager for Micro-Businesses



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FINAL YEAR PROJECT

PROPOSAL

TOPIC:

Biz-ight

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Introduction

The software engineering Final Year Project Biz-ight aims to create a decision support and automation dashboard which will support micro and home-based businesses. The project helps small business owners to control their sales operations and inventory management and pricing strategies and expense tracking while enabling them to make decisions based on data analysis.

The project requires students to create an innovative software system that solves real-world problems which micro-entrepreneurs face when they operate their businesses through social media platforms and basic e-commerce websites.

These businesses often rely on manual record-keeping and intuition-based decisions, which leads to inefficiencies and financial losses. Biz-ight combines operational e-commerce system data with its decision support system and automation processes and AI-based explanations to generate operational insights.

Objective

The main goals of this project are:

- To create a web-based decision support dashboard which will assist micro-businesses in their business operations
- To achieve automatic sales and inventory and expense monitoring through system integration
- To deliver rule-based business insights which will guide pricing and inventory control and expense tracking
- To enable decision-making through AI-based summary and explanation systems
- To test system performance in decreasing manual tasks while enhancing business decision-making processes.

Background/Existing System

Micro-businesses depend on social media networks and basic e-commerce platforms to operate their sales activities. The platforms enable users to perform transactions but they mainly concentrate on processing orders and maintaining records instead of assisting with business analysis and decision-making tasks.

Businesses encounter difficulties because existing inventory management and accounting software solutions:

- Present excessive technical difficulties which non-technical users cannot handle
- Targeting enterprise customers
- Provide static reports which do not deliver operational recommendations

Business owners encounter difficulties because they need to deal with:

- Unreliable methods for establishing product prices
- Insufficient methods for managing their stock. Their operational costs which they must manage remain hidden
- They make decisions based on immediate requirements instead of planning ahead

Small business owners show increasing interest in simplified automated intelligent systems which deliver business functions dedicated to their needs according to market patterns. The available solutions fail to support decision-making processes because they either use non-transparent artificial intelligence systems or they do not present accessible decision-making solutions to small business operations. The integrated platform unites rule-based decision support with workflow automation and AI-based explanation capabilities to solve existing system deficiencies.

Proposed Solution

The proposed solution is a **decision support and automation system** integrated with an e-commerce website. The system collects sales and inventory data through APIs or webhooks and processes it using predefined business rules.

The solution follows a layered approach:

Data Integration Layer

- Receives sales events from the e-commerce system
- Updates inventory and transaction records

Decision Support Layer

These KPIs form the basis of rule-based recommendations.

Key Performance Indicators:

- Revenue Growth Rate (RGR)
 $RGR = (\text{Revenue_current} - \text{Revenue_previous}) / \text{Revenue_previous}$
- Inventory Turnover Ratio (ITR)
 $ITR = \text{Cost of Goods Sold} / \text{Average Inventory}$
- Expense Ratio (ER)
 $ER = \text{Total Expenses} / \text{Total Revenue}$
- Stock Coverage Period (SCP)
 $SCP = \text{Current Stock} / \text{Daily Sales Rate}$

Rule-Based Evaluation Examples

The system applies deterministic rule conditions such as:

- If Current Stock < Reorder Threshold → Low Stock Alert
- If Stock Coverage Period < 3 days → Urgent Reorder Recommendation
- If Inventory Turnover Ratio < 1.5 → Slow-Moving Inventory Warning
- If Expense Ratio > 0.6 → Expense Spike Warning
- If Revenue Growth Rate < -0.1 → Sales Decline Alert

Threshold values are configurable and may be adjusted based on business category.

Business Health Score

The overall business performance is aggregated into a Business Health Score (BHS):

$$BHS = 0.4(ITR_{norm}) + 0.35(RGR_{norm}) + 0.25(1 - ER_{norm})$$

All KPI values are normalized between 0 and 1 before aggregation to ensure balanced contribution to the final score.

AI-Assisted Explanation Layer:

The AI-Assisted Explanation Layer does not make business decisions. The rule-based Decision Support Engine generates all decision outcomes. This layer exists to transform structured system outputs into understandable explanations that non-technical business users can comprehend.

The AI module receives predefined KPI values and triggered rule outputs from the decision engine. Structured prompt templates are used to develop explanations that match system-calculated metrics while avoiding any external assumptions.

The Appendix section contains a system block diagram which shows how data flows between different system components.

Project Scope

In-Scope

- Integration with an e-commerce website for sales data collection
- Inventory and expense management
- Rule-based pricing and inventory decision support
- Business health score calculation
- Automation of alerts and reports
- AI-assisted explanation of system recommendations

Out-of-Scope

- Payment processing
- Logistics and shipping management
- Tax filing and legal compliance
- Multi-vendor marketplace support
- Native mobile application development

Project Features

Admin / Business Owner Role

- Product and inventory management
- Sales and expense logging
- Dashboard visualization of KPIs
- Pricing and inventory recommendations
- Automated alerts (low stock, slow-moving items, expense spikes)
- Monthly business summary with AI-assisted explanation

A comprehensive use case model of the proposed system is included in the Appendix section.

Solution Application Areas

The proposed system is applicable in the following areas:

- Micro and home-based businesses
- Handmade crafts and jewelry businesses
- Small clothing brands
- Home-based food businesses
- Social media-driven e-commerce operations

Benefits

- Reduced administrative workload
- Improved pricing consistency
- Better inventory control
- Early detection of expense inefficiencies
- Accessible business insights for non-technical users

The project targets the small business and e-commerce software domain, addressing a practical and growing market need.

Tools/Technology

- Backend Framework: Django (Python)
- Database: PostgreSQL
- Frontend: Django Templates with Bootstrap 5
- Data Visualization: Chart.js
- Background Task Processing: Celery with Redis
- AI Integration: External AI API (REST-based)
- Version Control: Git & GitHub
- Development Environment: Visual Studio Code

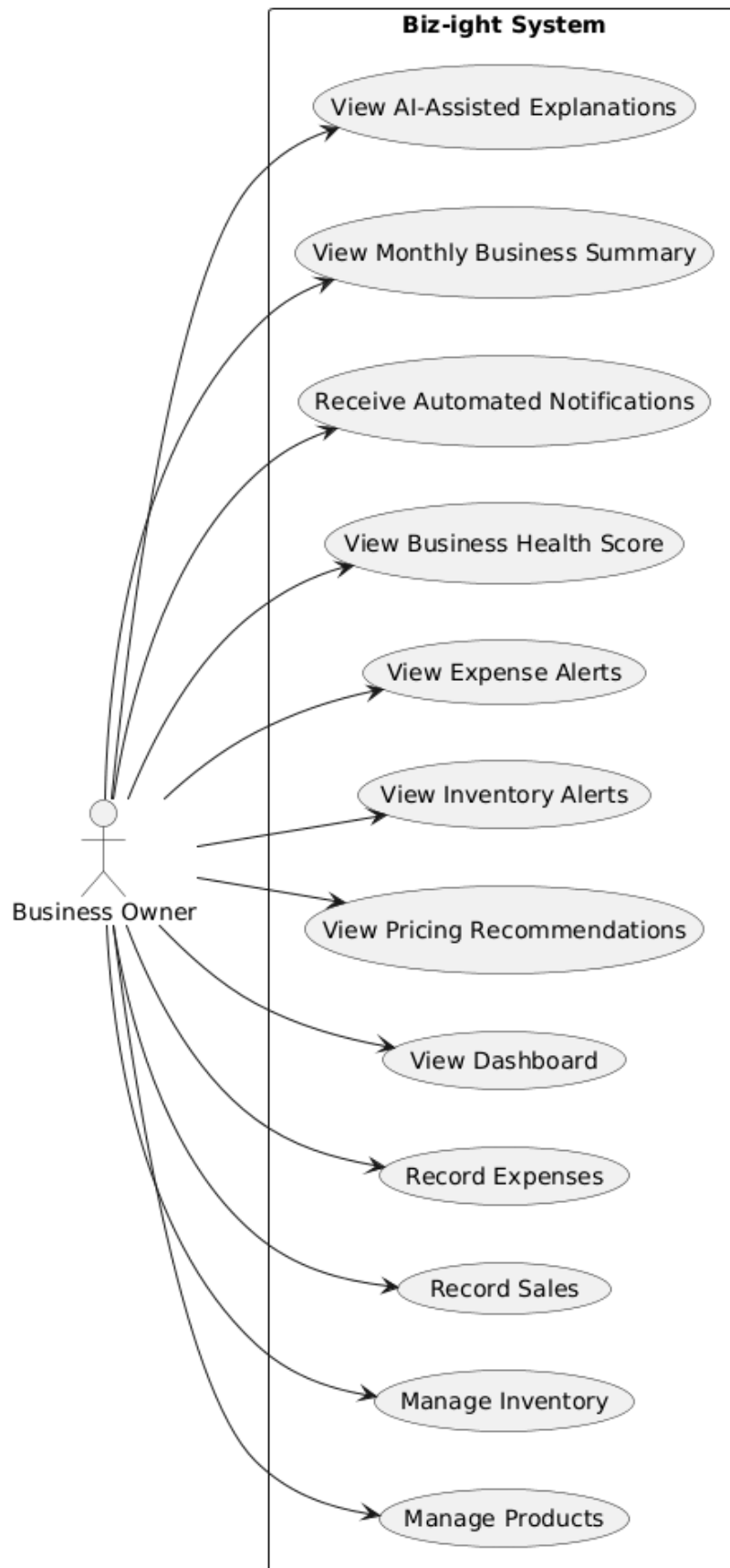
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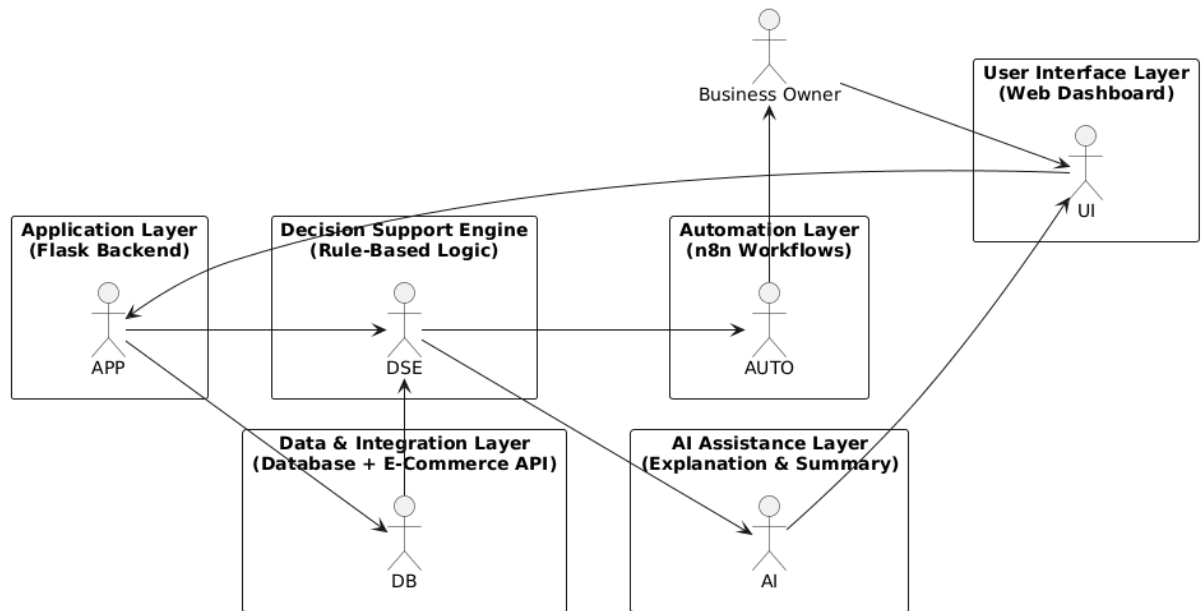
Appendices

- System architecture diagram
- Data flow diagram
- Use case diagram

USECASE DIAGRAM:



SYSTEM ARCHITECTURE DIAGRAM:



DATAFLOW DIAGRAM:

